

BEC303 — Network Analysis

EC branch, semester 3, 2025 scheme. Phase 1 output, generated by raaghu-pc.

Systematic methods for solving electrical networks — nodal and mesh analysis, network theorems, transient response, Laplace-domain analysis and two-port characterisation.

What is in this folder

Path	Contents
network-analysis.js	The machine-readable content file Phase 2 converts. 5 modules, 80 teaching units, quizzes and assignments.
ppt/	Five .pptx decks, 56 slides each, plus one .inspect.ndjson sidecar per deck.
quizzes/	Five quiz PDFs with answer keys, model answers and marking schemes, plus five Moodle GIFT files.
assignments/	Five assignment PDFs with full four-level rubrics and CO attainment tables.
pyq/	Copies of the branch-level merged question papers and PYQ_Index.pdf.
syllabus/	The syllabus extract, a copy of the source syllabus PDF, and Syllabus_Mapping.pdf.
MANIFEST.json	Counts, provenance and the result of every verification check.

Deck conventions Phase 2 depends on

- Every text shape carries a name: `title`, `objective` or `text`. There are no unnamed text shapes.
- **No PowerPoint animation or slide transition XML is present.** Animation is written as an `Animation instructions:` line in the speaker notes, and Phase 2 rebuilds the motion in React from that line.
- Every slide has a notes block with all six headings in this order: `What to say:`, `Pause cue:`, `Pointing cue:`, `Question to ask students:`, `Animation instructions:`, `Transition style:`, followed by a `[Sources]` line.
- Each unit in the content file carries a `slides` array naming the deck slides that teach it. That array is the join key between the deck and the content file.
- Each unit carries a `visual id` and a `visualSpec`. The `visualSpec` reveal order and the matching slide's `Animation instructions:` line describe the same sequence.

Module index

Module	Title	Hours	Units	Slides	Chapters
1	Basic Concepts — Mesh and Node Analysis	15	16	56	Hayt 8e Ch. 1 (1.1), Ch. 2 (2.3), Ch. 3 (3.2–3.3), Ch. 4 (4.1–4.4), Ch. 5 (5.2, 5.5), Ch. 10 (10.1–10.6), pp. 2–397
2	Network Theorems for DC and AC Circuits	13	16	56	Hayt 8e Ch. 5 (5.1, 5.3, 5.4), Ch. 10 (10.7), pp. 123–154 and 397–406; reciprocity from syllabus TB2 (Channa Venkatesh & Ganesh Rao) §4.9

Module	Title	Hours	Units	Slides	Chapters
3	Transient Behavior and Initial Conditions	14	16	56	Hayt 8e Ch. 7 (7.1, 7.2, 7.7), Ch. 8 (8.1–8.9), Ch. 9 (9.1–9.7), Ch. 10 (10.1–10.7), pp. 217–247, 261–361 and 371–406
4	Laplace Transformation and Applications	14	16	56	Hayt 8e Ch. 14 (14.3–14.7), Ch. 15 (15.1–15.8), pp. 540–564 and 571–610; waveform and network synthesis from syllabus TB2 §6.8
5	Two-Port Networks	14	16	56	Hayt 8e Ch. 17 (17.1–17.6), pp. 687–720

Sources

- Syllabus: [3ecsy11.pdf](#), VTU EC semester 3, 2025 scheme.
- Prescribed textbook 1: W. H. Hayt, J. E. Kemmerly and S. M. Durbin, *Engineering Circuit Analysis*, McGraw-Hill / Tata McGraw-Hill, 8th edition, 2012. File: [William_Hart_Hayt_Jack_E_Kemmerly_Steven_M_Durbz-lib.org_.pdf](#). Machine-extractable: yes, roughly 2.0 million characters; all TB1 chapter and page references come from the book's own table of contents.
- Prescribed TB2 as listed by the syllabus: K. Channa Venkatesh and D. Ganesh Rao, *Network Analysis - A Simplified Approach*, Cengage India, 2nd edition, 2012. **Not supplied in this environment.** Used by the syllabus for Module 2 (reciprocity, §4.9) and Module 4 (waveform and network synthesis, §6.8). Those units are aligned to the verbatim syllabus headings, cite only the chapter numbers the syllabus gives, and say so in their [Sources] note.
- Previous years' papers: [public/Previous Years QPs/EC/](#), copied unchanged. See [pyq/PYQ_Index.pdf](#) for the scheme-drift warning — Network Analysis is examined under BEC304 in those papers, not BEC303.

Regenerating

```
node temp/.tools/build-decks.mjs <subjectDir> network-analysis.js
node temp/.tools/build-docs.mjs <subjectDir> network-analysis.js <generatedBy> <pyqDir>
<syllabusPdf>
node temp/.tools/verify.mjs <subjectDir> network-analysis.js
```

Generator scripts and `node_modules` live in `temp/.tools/`. Nothing outside `temp/` was written.